SUPERMARKET SALES ANALYSIS

Business questions:-

-What factors influence supermarket sales and profitability across branches?

-How do customer type, product line, and time affect revenue and satisfaction?

Exploratory analysis:

-Which city generates the highest revenue?

-Which branch performs best?

-What is the total sales over time (daily/monthly)?

-What time of day has the highest sales?

Which product line sells the most?

-Which product line generates the highest profit (gross income)?

-Which products have low sales but high price?

-Do members vs normal customers spend differently?

-Which gender spends more?

-What is the average quantity per customer type?

-What is the average rating per branch?

-Which product line has the lowest ratings?

-Is there a relationship between rating and sales?

-Which payment method is most used?

-Do people who use certain payment methods spend more?

-Which branch has the highest gross income?

-How does gross margin percentage vary?

-What factors affect profit most (price, quantity, tax

```
from google.colab import files
Uploaded = files.upload()
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Choose files supermarket_sales.csv

supermarket_sales.csv(text/csv) - 131528 bytes, last modified: 23/03/2026 - 100% done

```
Saving supermarket_sales.csv to supermarket_sales.csv
```

```
data = pd.read_csv("supermarket_sales.csv")
data.head()
```

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity
0	750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	7
1	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	5
2	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	7
3	123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	8
4	373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	7

Next steps: Generate code with data New interactive sheet

```
data.columns
```

```
Index(['Invoice ID', 'Branch', 'City', 'Customer type', 'Gender',
       'Product line', 'Unit price', 'Quantity', 'Tax 5%', 'Total',
'Date',
       'Time', 'Payment', 'cogs', 'gross margin percentage', 'gross
income',
       'Rating'],
      dtype='object')
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 17 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   Invoice ID               1000 non-null   object
 1   Branch                   1000 non-null   object
 2   City                     1000 non-null   object
```

```
 3   Customer type            1000 non-null   object
 4   Gender                   1000 non-null   object
 5   Product line             1000 non-null   object
 6   Unit price               1000 non-null   float64
 7   Quantity                 1000 non-null   int64
 8   Tax 5%                   1000 non-null   float64
 9   Total                    1000 non-null   float64
 10  Date                     1000 non-null   object
 11  Time                     1000 non-null   object
 12  Payment                  1000 non-null   object
 13  cogs                     1000 non-null   float64
 14  gross margin percentage  1000 non-null   float64
 15  gross income             1000 non-null   float64
 16  Rating                   1000 non-null   float64
dtypes: float64(7), int64(1), object(9)
memory usage: 132.9+ KB
```

```
data.describe()
```

	Unit price	Quantity	Tax 5%	Total	cogs	gros margi percentag
count	1000.000000	1000.000000	1000.000000	1000.000000	1000.00000	1.000000e+0
mean	55.672130	5.510000	15.379369	322.966749	307.58738	4.761905e+0
std	26.494628	2.923431	11.708825	245.885335	234.17651	6.131498e-1
min	10.080000	1.000000	0.508500	10.678500	10.17000	4.761905e+0
25%	32.875000	3.000000	5.924875	124.422375	118.49750	4.761905e+0
50%	55.230000	5.000000	12.088000	253.848000	241.76000	4.761905e+0
75%	77.935000	8.000000	22.445250	471.350250	448.90500	4.761905e+0
max	99.960000	10.000000	49.650000	1042.650000	993.00000	4.761905e+0

```
data.shape
```

```
(1000, 17)
```

```
data.isnull().sum()
```

	0
Invoice ID	0
Branch	0
City	0
Customer type	0
Gender	0
Product line	0
Unit price	0
Quantity	0
Tax 5%	0
Total	0
Date	0
Time	0
Payment	0
cogs	0
gross margin percentage	0
gross income	0
Rating	0

dtype: int64

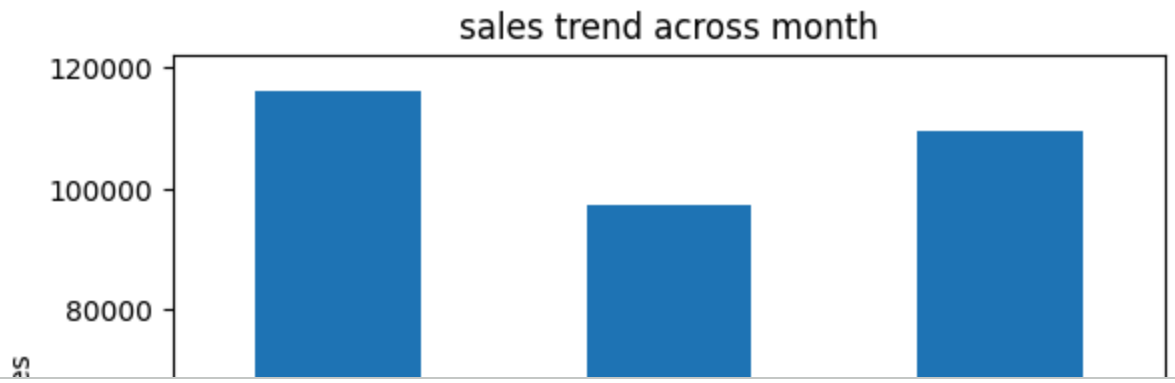
```
#change the the date to datetime
data['Date'] = pd.to_datetime(data['Date'])
data['Month'] = pd.to_datetime(data['Date']).dt.month
data['Day'] = data['Date'].dt.day
data["hours"] = data["Date"].dt.hour
```

```
month_sales = data.groupby(["Month"])["Total"].sum()
print(month_sales)
```

```
Month
1    116291.868
2     97219.374
3    109455.507
Name: Total, dtype: float64
```

```
month_sales. plot(kind="bar")
plt.title('sales trend across month')
```

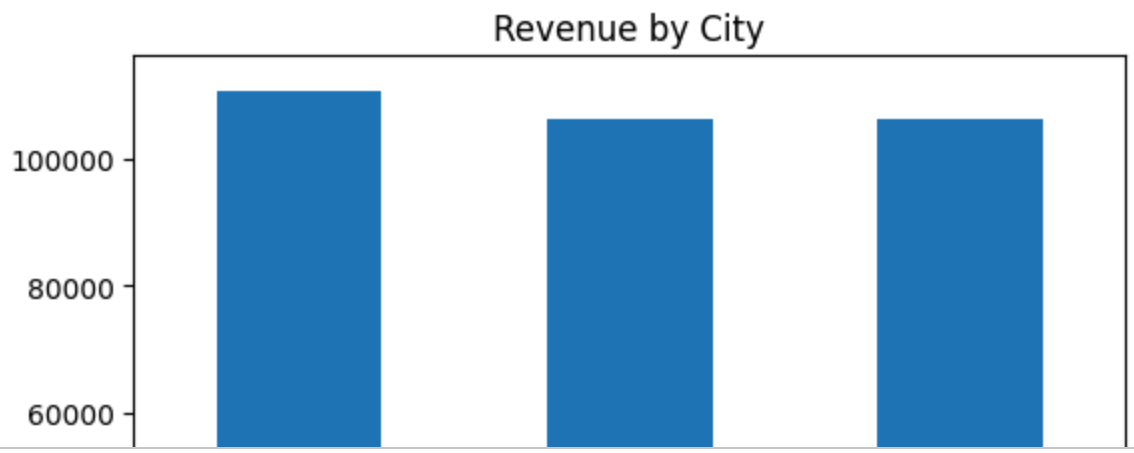
```
plt.xlabel('month')
plt.ylabel('total sales')
plt.xticks(size=8)
plt.show()
```




```
#city with highest sales
city_sales = data.groupby('City')['Total'].sum().sort_values(ascending=False)
print(city_sales)

city_sales.plot(kind='bar')
plt.title('Revenue by City')
plt.show()
```

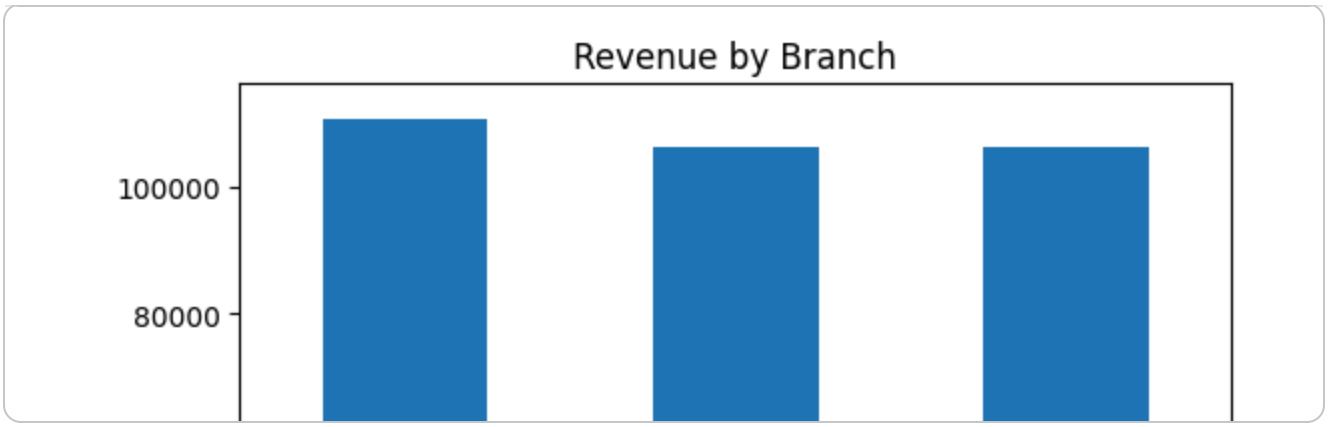
```
City
Naypyitaw    110568.7065
Yangon       106200.3705
Mandalay     106197.6720
Name: Total, dtype: float64
```



Insight

Naypyitaw city has highest sale followed by yangon then mandalay

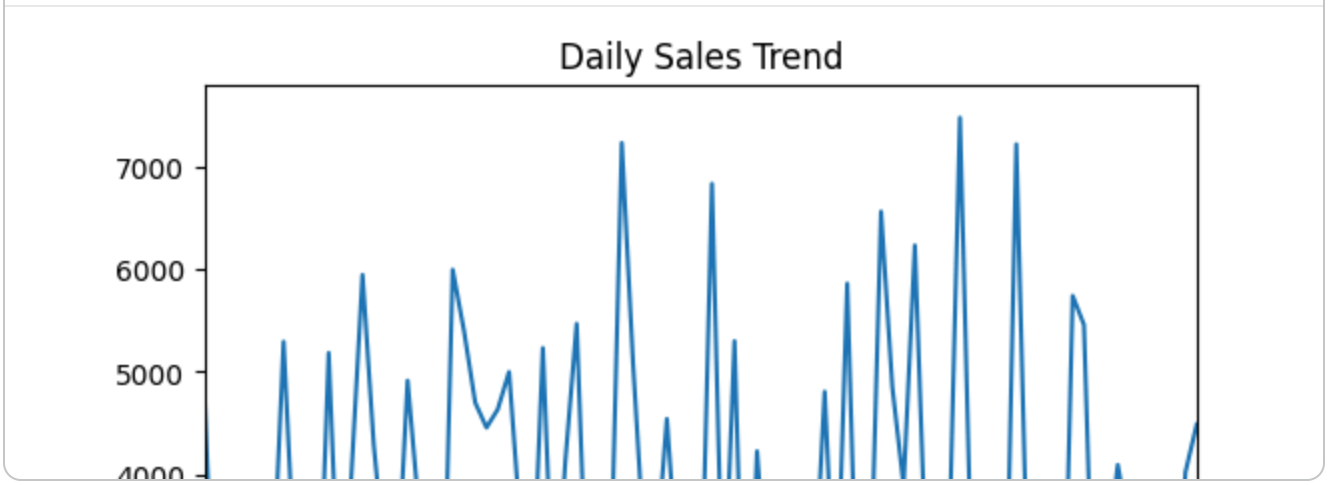
```
#which branch has highest sales
branch_sales = data.groupby('Branch')['Total'].sum().sort_values(ascending=Fals
branch_sales.plot(kind='bar')
plt.title('Revenue by Branch')
plt.xticks(rotation='vertical')
plt.show()
```



Insight

Branch C has highest revenues followed by branch A then B

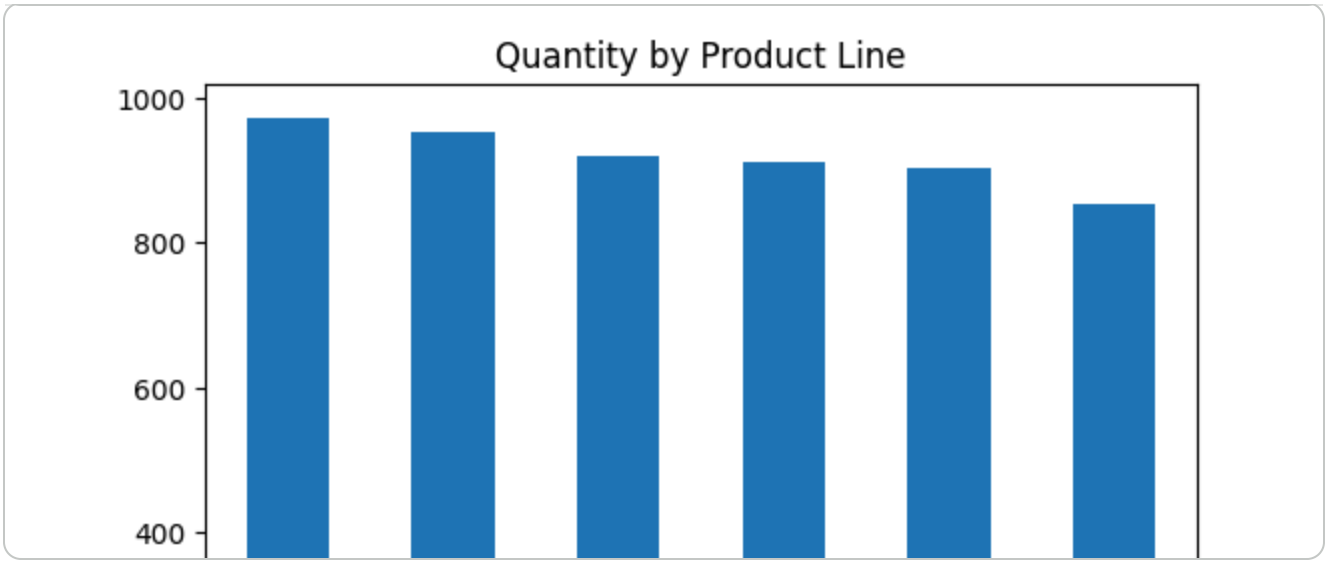
```
#trend of sales over time
daily_sales = data.groupby('Date')['Total'].sum()
daily_sales.plot()
plt.title('Daily Sales Trend')
plt.show()
```



Insight

Daily sales vary each other across time.

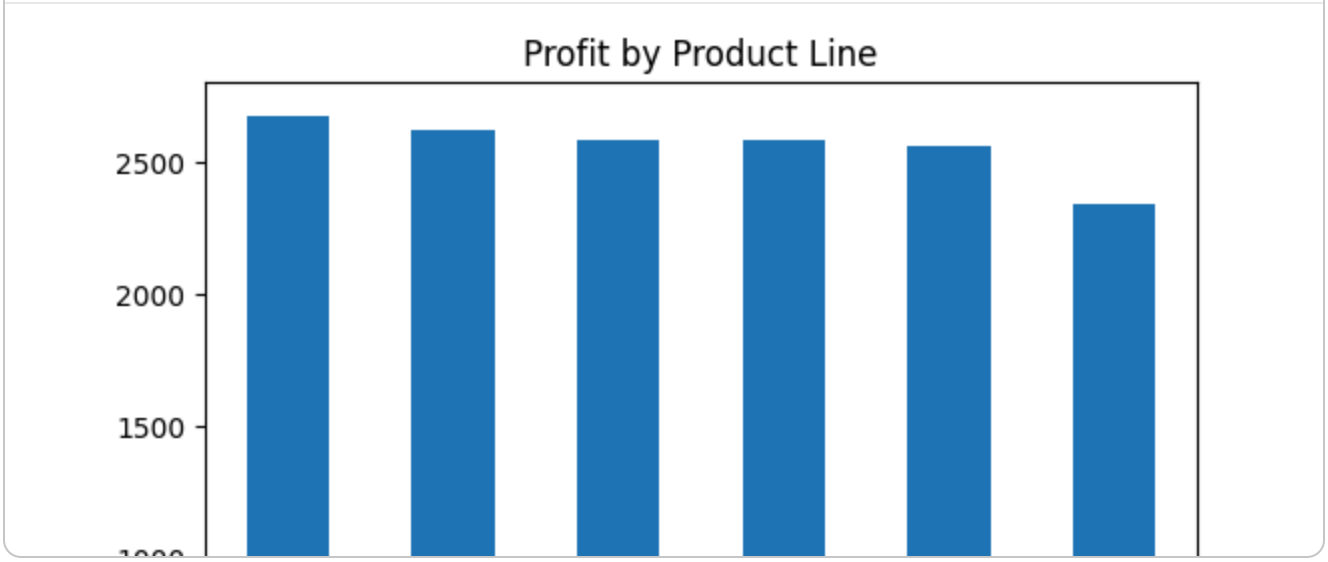
```
#best selling product
product_sales = data.groupby('Product line')['Quantity'].sum().sort_values(asc
product_sales.plot(kind='bar')
plt.title('Quantity by Product Line')
plt.show()
```



Insight

Electronic accessories are the best selling product.

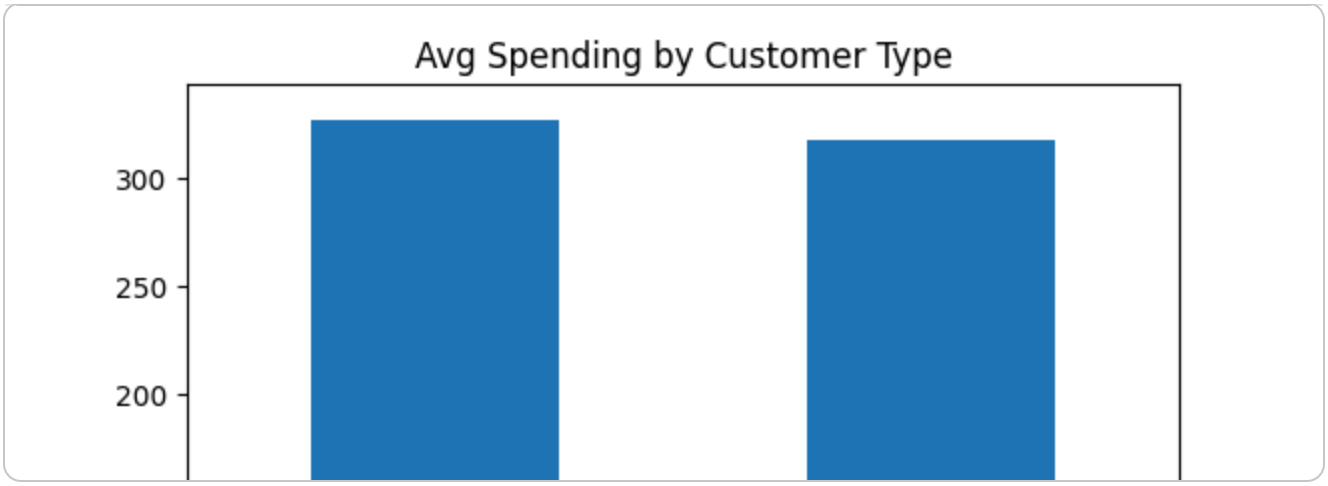
```
#Product with high revenue
profit_product = data.groupby('Product line')['gross income'].sum().sort_values
profit_product.plot(kind='bar')
plt.title('Profit by Product Line')
plt.show()
```



Insight

Food and beverages product produce high revenue.

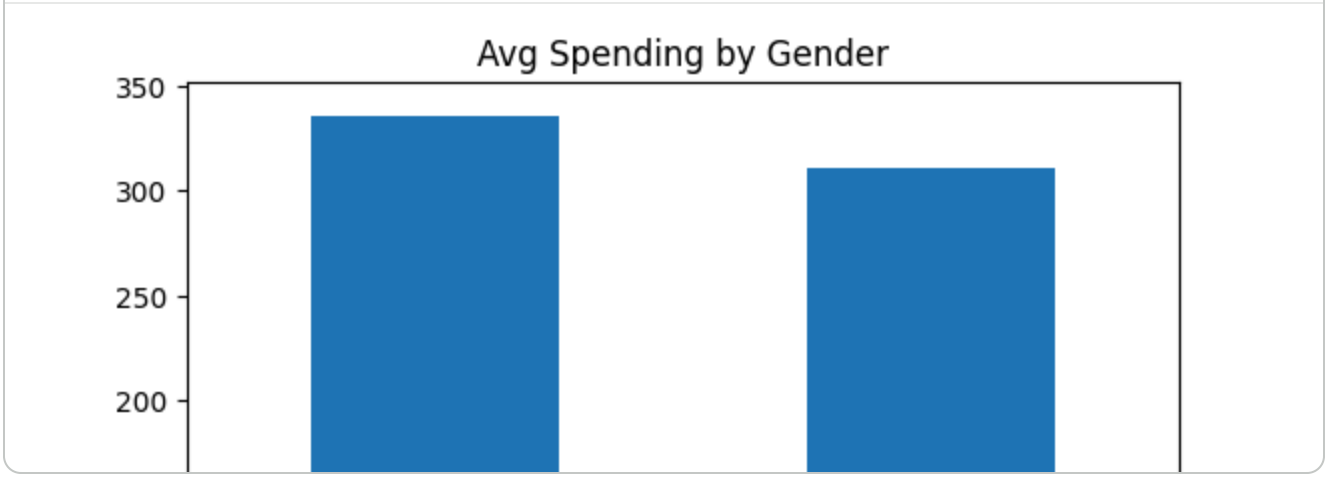
```
#which customer buy more
customer_spending = data.groupby('Customer type')['Total'].mean()
customer_spending.plot(kind='bar')
plt.title('Avg Spending by Customer Type')
plt.show()
```



Insight

Member buy more than normal customer.

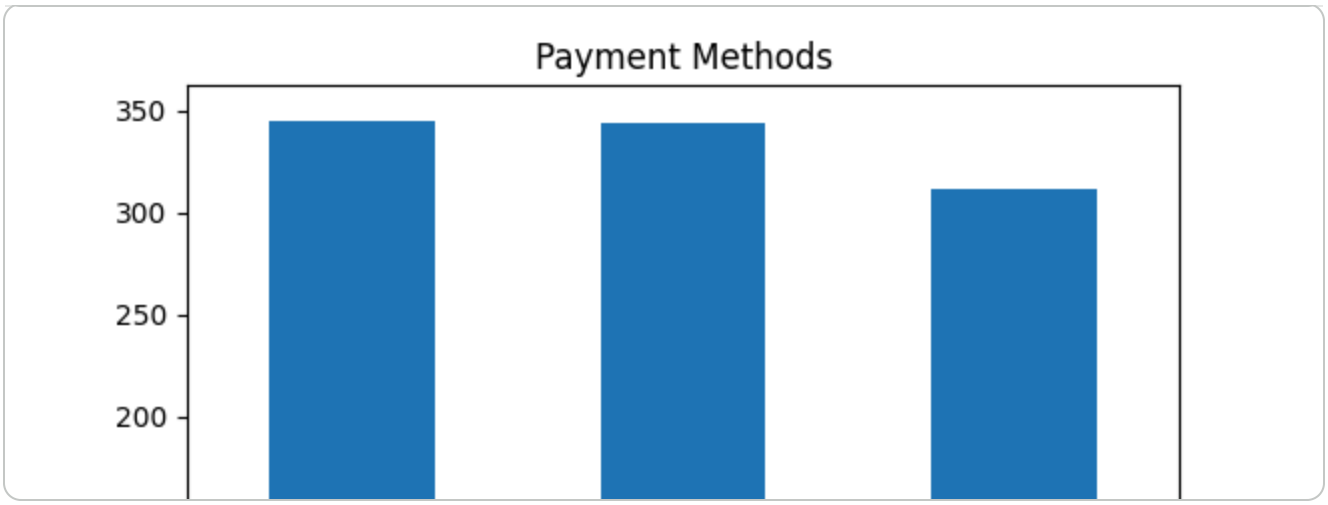
```
#which gender spending more
gender_spending = data.groupby('Gender')['Total'].mean()
gender_spending.plot(kind='bar')
plt.title('Avg Spending by Gender')
plt.show()
```



Insight

Females buy more than male

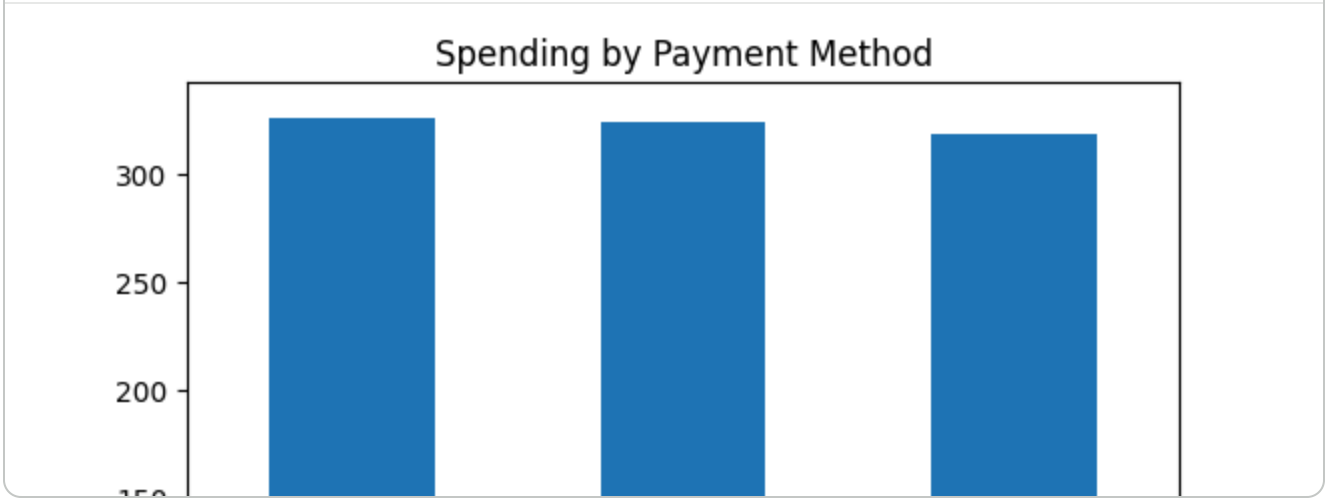
```
#most used payment method
payment_count = data['Payment'].value_counts()
payment_count.plot(kind='bar')
plt.title('Payment Methods')
plt.show()
```



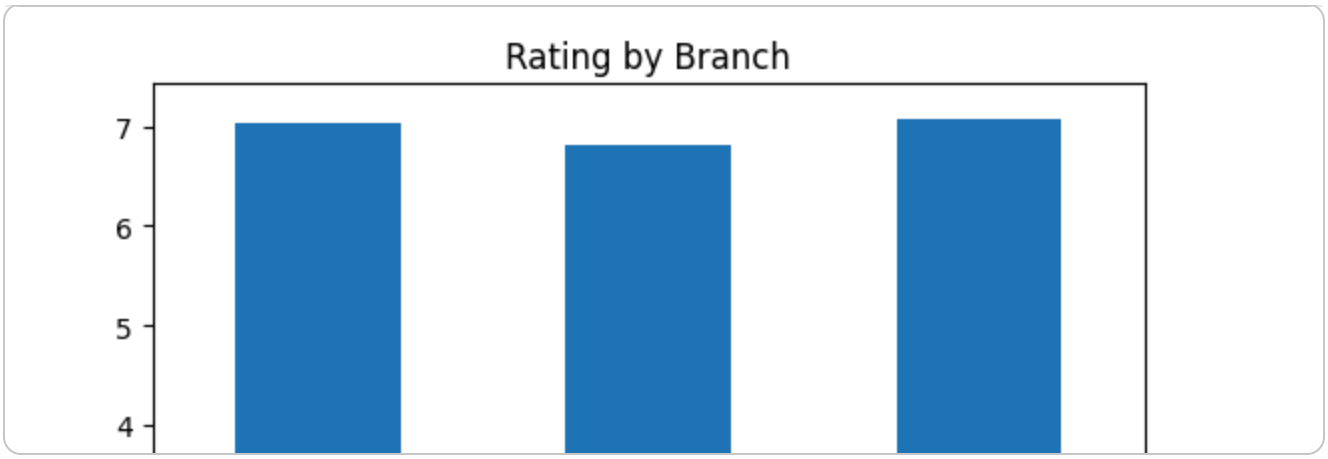
Insight

Ewallet and cash are most used payment method

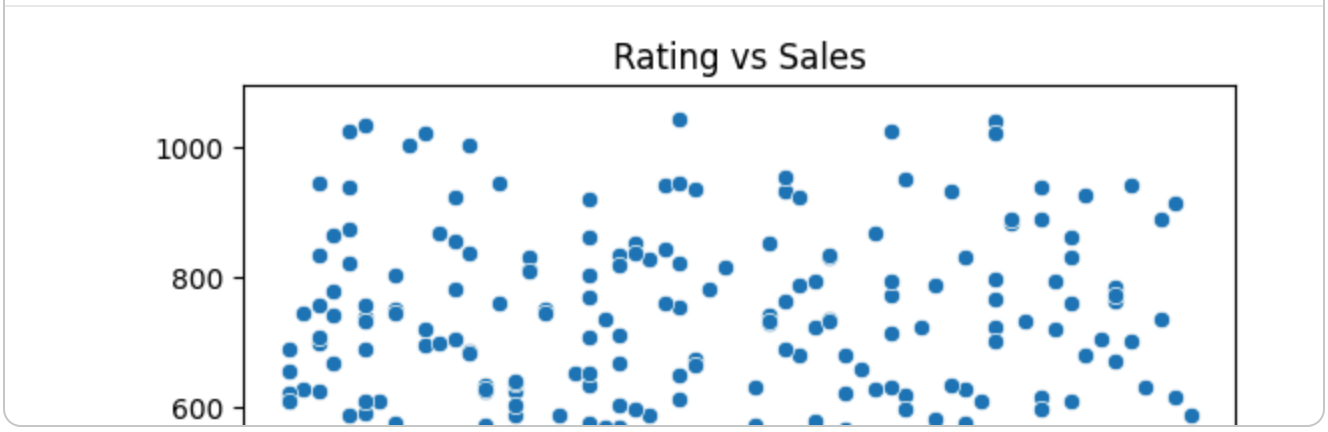
```
#spending
payment_spending = data.groupby('Payment')['Total'].mean()
payment_spending.plot(kind='bar')
plt.title('Spending by Payment Method')
plt.show()
```



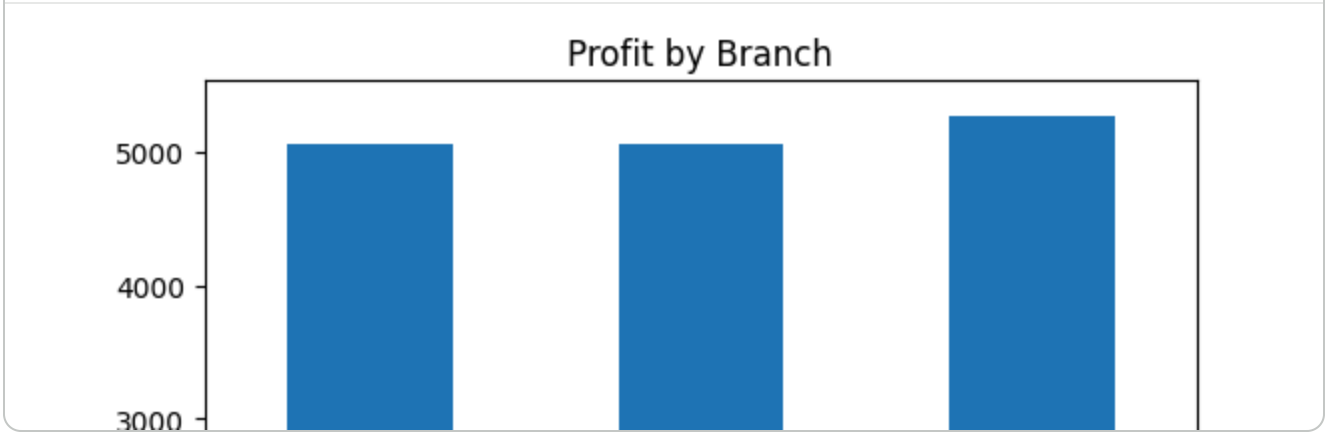

```
#rating per branch
rating_branch = data.groupby('Branch')['Rating'].mean()
rating_branch.plot(kind='bar')
plt.title('Rating by Branch')
plt.show()
```



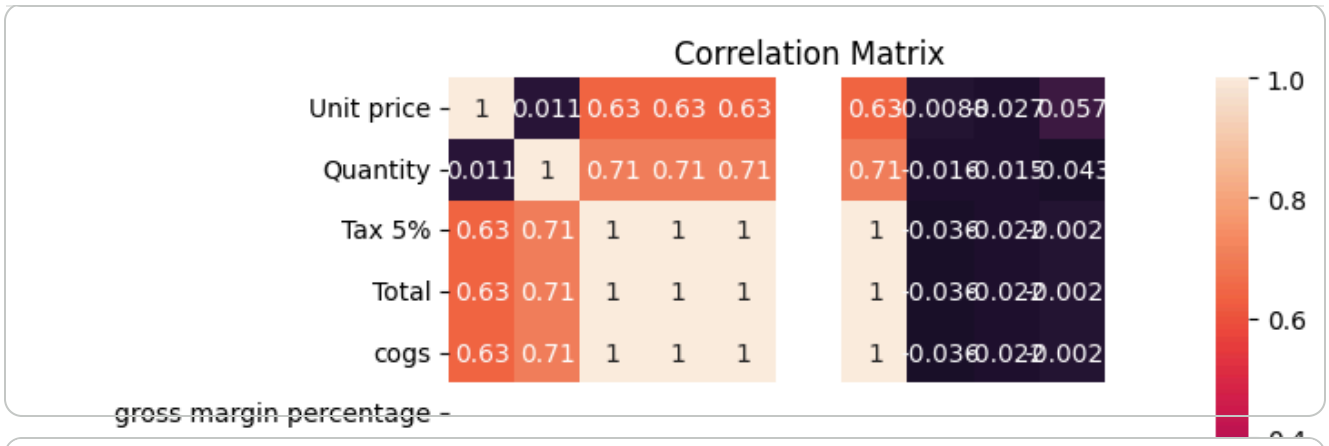

```
#rating
sns.scatterplot(x='Rating', y='Total', data=data)
plt.title('Rating vs Sales')
plt.show()
```

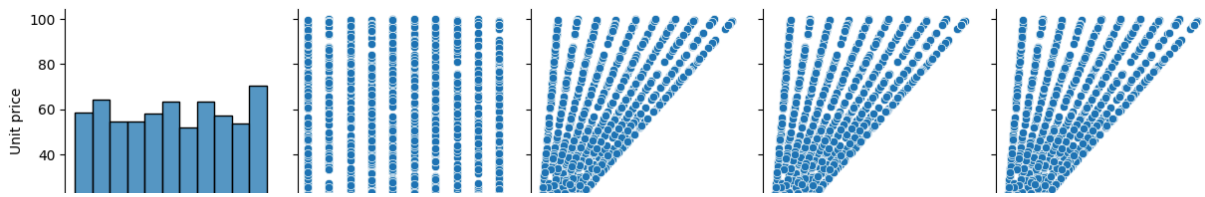



```
#profit by branch
profit_branch = data.groupby('Branch')['gross income'].sum()
profit_branch.plot(kind='bar')
plt.title('Profit by Branch')
plt.show()
```




```
#correlation
sns.heatmap(data.corr(numeric_only=True), annot=True)
plt.title('Correlation Matrix')
plt.show()
```




```
features = ['Unit price', 'Quantity', 'Tax 5%', 'cogs', 'gross income']
sns.pairplot(data[features])
plt.show()
```